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Chapter 11 - The Method Of Undetermined Coefficients

Chapter 10 solved homogeneous constant-coefficient equations of any finite order. This chapter adds a forcing term and asks a new question:

once the complementary solution is known, how can we construct one function that produces the required forcing when substituted into the differential equation?

The method of undetermined coefficients answers that question for a useful class of forcing functions. Its central strategy is to predict the **shape** of a particular solution before calculating its numerical coefficients.

The main questions are:

- why does one particular solution complete the general solution?
- how does the forcing term determine the trial family?
- why must polynomial trials contain every lower power?
- why must sine and cosine be included together?
- how are product and sum forcing terms handled?
- what changes when the first trial overlaps the homogeneous solution?
- when is this method unavailable?

From A Homogeneous Family To A Forced Solution

The Goal Of This Block

The goal is to understand the structure:

$$y = y_h + y_p$$

before learning how to construct y_p .

Here:

- y_h is the general solution of the associated homogeneous equation
- y_p is one particular solution of the nonhomogeneous equation

The key distinction is that y_h contains arbitrary constants, while y_p does not.

The Operator View Of A Forced Equation

Write an n th-order linear constant-coefficient equation as:

$$L[y] = g(x),$$

where:

- L is the linear differential operator on the left
- $y = y(x)$ is the unknown function
- $g(x)$ is the forcing function on the right

For example:

$$y'' - 3y' + 2y = 5x$$

The equation $y'' - 3y' + 2y = 5x$ can be written as:

$$L[y] = 5x,$$

with:

$$L[y] = y'' - 3y' + 2y.$$

The **associated homogeneous equation** is obtained by replacing the forcing with zero:

$$L[y] = 0.$$

This does not mean that the original forcing disappears from the problem. It creates a related equation whose solution family supplies the complementary part y_h .

Why $y_h + y_p$ Works

Suppose:

$$L[y_h] = 0$$

and:

$$L[y_p] = g(x).$$

Because L is linear:

$$L[y_h + y_p] = L[y_h] + L[y_p].$$

Substitute the two known outputs:

$$\begin{aligned} L[y_h + y_p] &= L[y_h] + L[y_p] \\ &= 0 + g(x) \\ &= g(x). \end{aligned}$$

Therefore $y_h + y_p$ satisfies the nonhomogeneous equation.

A useful interpretation:

y_h contains every freely adjustable motion allowed by the homogeneous equation, while y_p supplies one response forced by the right side.

Why One Particular Solution Is Enough

Suppose $y_{p,1}$ and $y_{p,2}$ are two particular solutions of the same equation:

$$L[y_{p,1}] = g(x), \quad L[y_{p,2}] = g(x).$$

Subtract the second equation from the first. By linearity:

$$\begin{aligned} L[y_{p,1} - y_{p,2}] &= L[y_{p,1}] - L[y_{p,2}] \\ &= g(x) - g(x) \\ &= 0. \end{aligned}$$

Thus $y_{p,1} - y_{p,2}$ is a homogeneous solution. Any difference between two particular solutions is already absorbed by the arbitrary constants in y_h .

That is why the method seeks **one** convenient y_p , not every possible particular solution.

A Small Structural Example

Consider:

$$y'' - y' - 6y = 12.$$

The associated homogeneous equation is:

$$y'' - y' - 6y = 0.$$

Its characteristic equation is:

$$r^2 - r - 6 = 0.$$

Factor:

$$(r - 3)(r + 2) = 0.$$

Therefore:

$$y_h = c_1 e^{3x} + c_2 e^{-2x}.$$

Now look for a constant particular solution:

$$y_p = A.$$

Because A is constant:

$$y'_p = 0, \quad y''_p = 0.$$

For $y_p = A$, both derivatives are zero. Substitute $y_p = A$, $y'_p = 0$, and $y''_p = 0$ into the original equation:

$$\begin{aligned}y''_p - y'_p - 6y_p &= 0 - 0 - 6A \\ &= -6A.\end{aligned}$$

The required forcing is 12, so:

$$-6A = 12.$$

Divide by -6 :

$$A = -2.$$

Thus one particular solution is:

$$y_p = -2.$$

Combine the homogeneous and particular parts:

$$\boxed{y = c_1 e^{3x} + c_2 e^{-2x} - 2}.$$

The constants c_1, c_2 are arbitrary because they belong to y_h . The coefficient A was not arbitrary: substitution forced it to equal -2 .

Block 1 Summary

For a linear nonhomogeneous equation $L[y] = g(x)$, the general solution has the form:

$$y = y_h + y_p.$$

The two parts have different jobs:

$$L[y_h] = 0, \quad L[y_p] = g(x).$$

Only one particular solution is required because the difference between any two particular solutions lies in the homogeneous family.

Reading The Forcing Fingerprint

The Goal Of This Block

The goal is to choose the correct trial family before calculating any coefficients.

The organising idea is:

include every function that can appear when the forcing function is differentiated repeatedly.

The trial is not a guess at the final numerical answer. It is a finite family with unknown coefficients that is broad enough to survive differentiation.

Why Derivative Closure Matters

The operator L combines a proposed function and several of its derivatives. If those derivatives leave the proposed family, substitution creates terms that the trial cannot represent.

For example, differentiating x^2 gives:

$$x^2 \rightarrow 2x \rightarrow 2 \rightarrow 0.$$

The finite family involved is:

$$\{x^2, x, 1\}.$$

Therefore a second-degree polynomial forcing requires the complete trial:

$$Ax^2 + Bx + C,$$

not merely Ax^2 .

Similarly:

$$\frac{d}{dx} \sin(\omega x) = \omega \cos(\omega x)$$

and:

$$\frac{d}{dx} \cos(\omega x) = -\omega \sin(\omega x).$$

Sine and cosine generate each other. A trial for either one must normally contain both.

The Basic Trial Ledger

Before checking for overlap with y_h , use the following base trials.

Forcing $g(x)$	Base trial for y_p
$P_n(x)$	$A_n x^n + A_{n-1} x^{n-1} + \cdots + A_1 x + A_0$
Ke^{ax}	Ae^{ax}
$K_1 \cos(\omega x) + K_2 \sin(\omega x)$	$A \cos(\omega x) + B \sin(\omega x)$

Here:

- $P_n(x)$ is a polynomial of degree n
- a and ω are known numbers from the forcing
- the capital letters A_j, A, B are coefficients to be determined

The values multiplying the forcing do not determine the trial's shape. For example, $19e^{-4x}$ and $-2e^{-4x}$ both lead to the base trial Ae^{-4x} .

Polynomial Forcing: Keep Every Lower Power

Suppose:

$$g(x) = 7x^3 - 2x.$$

Although the x^2 and constant terms are absent, the forcing has degree 3. The complete base trial is:

$$y_p = Ax^3 + Bx^2 + Cx + D.$$

The missing powers cannot be omitted. Differentiation and the lower-order terms in L can create contributions at those powers even when their target coefficients are zero.

Coefficient comparison will later include:

$$0x^2 \quad \text{and} \quad 0.$$

Those zero targets are genuine equations for the unknown trial coefficients.

Trigonometric Forcing: Keep Both Partners

Suppose:

$$g(x) = 9 \sin(5x).$$

The forcing visibly contains only sine, but its derivative contains cosine. Therefore the base trial is:

$$y_p = A \cos(5x) + B \sin(5x).$$

If the forcing were $9 \cos(5x)$, the same two-function trial would be used. One of A or B may turn out to be zero after substitution, but that is a conclusion of the calculation rather than an assumption made beforehand.

Products Inherit Every Factor

Suppose the forcing is:

$$g(x) = e^{-2x}(4x^2 + 1).$$

This is an exponential multiplied by a second-degree polynomial. Preserve the exponential and attach a complete second-degree polynomial:

$$y_p = e^{-2x}(Ax^2 + Bx + C).$$

For a polynomial multiplied by a trigonometric function, both trigonometric partners need polynomial coefficients. For example:

$$g(x) = (x - 3) \cos(5x)$$

leads to:

$$y_p = (Ax + B) \cos(5x) + (Cx + D) \sin(5x).$$

The two polynomials have the same degree as the polynomial in the forcing.

Sums Produce Sums Of Trials

Suppose:

$$g(x) = 2e^x + x^2 \sin(3x).$$

Treat the two forcing components separately.

For $2e^x$, use:

$$y_{p,1} = Ae^x.$$

For $x^2 \sin(3x)$, use a complete quadratic on both trigonometric partners:

$$y_{p,2} = (Bx^2 + Cx + D) \cos(3x) + (Ex^2 + Fx + G) \sin(3x).$$

Add the two trial components:

$$\begin{aligned} y_p = & Ae^x \\ & + (Bx^2 + Cx + D) \cos(3x) \\ & + (Ex^2 + Fx + G) \sin(3x). \end{aligned}$$

This base trial is not yet final. It must still be compared with y_h for overlap.

The Trial-Selection Checklist

Before differentiating a trial, ask:

1. What basic forcing families are present?
2. What polynomial degree must be preserved?
3. Is an exponential factor present?
4. What trigonometric frequency must be preserved?
5. Have both sine and cosine been included?
6. If the forcing is a sum, has each component received its own trial?
7. Does any part of the combined trial overlap y_h ?

The seventh question is essential. Blocks 3 and 4 first study non-overlapping examples; Block 5 develops the correction for overlap.

Block 2 Summary

Trial selection is controlled by derivative closure. Preserve:

- every polynomial power down to the constant term
- the forcing exponential
- both sine and cosine at the forcing frequency
- every factor in a product
- every component in a sum

Then compare the complete base trial with the homogeneous solution before using it.

Solving The Three Basic Nonresonant Cases

The Goal Of This Block

The goal is to carry out the complete method for polynomial, exponential, and trigonometric forcing when the base trial does not overlap y_h .

Every example follows the same sequence:

1. solve the associated homogeneous equation
2. choose and check the base trial
3. differentiate the trial
4. substitute into the differential equation
5. compare like coefficients
6. solve for the trial coefficients
7. write $y = y_h + y_p$

Worked Polynomial Forcing Example

Solve:

$$y'' + y' - 6y = -12x^2 + 10x - 15.$$

Step 1: Solve the associated homogeneous equation. Replace the forcing by zero:

$$y'' + y' - 6y = 0.$$

The characteristic equation is:

$$r^2 + r - 6 = 0.$$

Factor:

$$(r - 2)(r + 3) = 0.$$

The roots are 2 and -3 , so:

$$y_h = c_1 e^{2x} + c_2 e^{-3x}.$$

Step 2: Build and check the trial. The forcing is a second-degree polynomial. Use the complete quadratic trial:

$$y_p = Ax^2 + Bx + C.$$

The homogeneous solution contains exponentials, not polynomial modes, so no overlap occurs.

Step 3: Differentiate the trial. Starting from $y_p = Ax^2 + Bx + C$:

$$y'_p = 2Ax + B,$$

$$y''_p = 2A.$$

Step 4: Substitute each expression. The left side is $y''_p + y'_p - 6y_p$. Substitute the trial and its derivatives:

$$y''_p + y'_p - 6y_p = 2A + (2Ax + B) - 6(Ax^2 + Bx + C).$$

Distribute -6 :

$$y''_p + y'_p - 6y_p = 2A + 2Ax + B - 6Ax^2 - 6Bx - 6C.$$

Group equal powers of x :

$$y''_p + y'_p - 6y_p = (-6A)x^2 + (2A - 6B)x + (2A + B - 6C).$$

Step 5: Match the forcing coefficient by coefficient. The differential equation requires:

$$(-6A)x^2 + (2A - 6B)x + (2A + B - 6C) = -12x^2 + 10x - 15.$$

Match the x^2 coefficients:

$$-6A = -12.$$

The current equation is $-6A = -12$. Divide both sides by -6 :

$$A = 2.$$

Using $A = 2$ from the x^2 coefficient, now match the x coefficients:

$$2A - 6B = 10.$$

Substitute $A = 2$:

$$4 - 6B = 10.$$

Subtract 4:

$$-6B = 6.$$

The current equation is $-6B = 6$. Divide both sides by -6 :

$$B = -1.$$

Match the constant coefficients:

$$2A + B - 6C = -15.$$

Substitute $A = 2$ and $B = -1$:

$$4 - 1 - 6C = -15.$$

Combine the known constants:

$$3 - 6C = -15.$$

Starting from $3 - 6C = -15$, subtract 3 from both sides:

$$-6C = -18.$$

Divide by -6 :

$$C = 3.$$

Step 6: Construct the particular and general solutions. Insert $A = 2$, $B = -1$, and $C = 3$ into the trial:

$$y_p = 2x^2 - x + 3.$$

Now combine y_h and y_p :

$$y = c_1 e^{2x} + c_2 e^{-3x} + 2x^2 - x + 3.$$

Worked Exponential Forcing Example

Solve:

$$y'' + 4y' + 3y = 30e^{2x}.$$

Step 1: Find the complementary solution. The associated characteristic equation is:

$$r^2 + 4r + 3 = 0.$$

Factor the characteristic equation $r^2 + 4r + 3 = 0$:

$$(r + 1)(r + 3) = 0.$$

Therefore:

$$y_h = c_1 e^{-x} + c_2 e^{-3x}.$$

Step 2: Choose and check the trial. The forcing is a constant multiple of e^{2x} , so the base trial is:

$$y_p = Ae^{2x}.$$

The homogeneous exponentials are e^{-x} and e^{-3x} . Since e^{2x} is not present in y_h , the base trial is valid without modification.

Step 3: Differentiate. Starting from $y_p = Ae^{2x}$:

$$y_p' = 2Ae^{2x},$$

$$y_p'' = 4Ae^{2x}.$$

Step 4: Substitute into the left side. Use $y_p'' + 4y_p' + 3y_p$:

$$\begin{aligned} y_p'' + 4y_p' + 3y_p &= 4Ae^{2x} + 4(2Ae^{2x}) + 3(Ae^{2x}) \\ &= 4Ae^{2x} + 8Ae^{2x} + 3Ae^{2x} \\ &= 15Ae^{2x}. \end{aligned}$$

The required forcing is $30e^{2x}$, so:

$$15Ae^{2x} = 30e^{2x}.$$

The exponential e^{2x} is nonzero, so divide both sides by it:

$$15A = 30.$$

Divide by 15:

$$A = 2.$$

Step 5: Write the complete solution. The particular solution is:

$$y_p = 2e^{2x}.$$

Therefore:

$$y = c_1e^{-x} + c_2e^{-3x} + 2e^{2x}.$$

Worked Trigonometric Forcing Example

Solve:

$$y'' + 2y' + 5y = -16\cos(3x) + 2\sin(3x).$$

Step 1: Find the complementary solution. The associated characteristic equation is:

$$r^2 + 2r + 5 = 0.$$

Complete the square:

$$(r + 1)^2 + 4 = 0.$$

From $(r + 1)^2 + 4 = 0$, the roots are:

$$r = -1 \pm 2i.$$

The complementary solution is:

$$y_h = e^{-x} (c_1 \cos(2x) + c_2 \sin(2x)).$$

Step 2: Choose and check the trial. The forcing frequency is 3, so include both trigonometric partners:

$$y_p = A \cos(3x) + B \sin(3x).$$

The homogeneous solution uses frequency 2 with an e^{-x} envelope. The trial uses frequency 3 with no exponential envelope, so no overlap occurs.

Step 3: Differentiate carefully. Starting from $y_p = A \cos(3x) + B \sin(3x)$:

$$y'_p = -3A \sin(3x) + 3B \cos(3x),$$

$$y''_p = -9A \cos(3x) - 9B \sin(3x).$$

Step 4: Substitute and group the two function types. Substitute into $y''_p + 2y'_p + 5y_p$:

$$\begin{aligned} y''_p + 2y'_p + 5y_p &= [-9A \cos(3x) - 9B \sin(3x)] \\ &\quad + 2[-3A \sin(3x) + 3B \cos(3x)] \\ &\quad + 5[A \cos(3x) + B \sin(3x)]. \end{aligned}$$

Collect the cosine terms:

$$(-9A + 6B + 5A) \cos(3x) = (-4A + 6B) \cos(3x).$$

Collect the sine terms:

$$(-9B - 6A + 5B) \sin(3x) = (-6A - 4B) \sin(3x).$$

If $L[y_{p,1}] = g_1(x)$ and $L[y_{p,2}] = g_2(x)$, linearity gives the combined particular solution:

$$\begin{aligned} y_p'' + 2y_p' + 5y_p &= (-4A + 6B) \cos(3x) \\ &\quad + (-6A - 4B) \sin(3x). \end{aligned}$$

Step 5: Match cosine and sine coefficients. The forcing is:

$$-16 \cos(3x) + 2 \sin(3x).$$

Therefore the coefficients satisfy:

$$\begin{aligned} -4A + 6B &= -16, \\ -6A - 4B &= 2. \end{aligned}$$

We now solve this system without skipping the elimination step. Multiply the first equation by 2:

$$-8A + 12B = -32.$$

Multiply the second equation by 3:

$$-18A - 12B = 6.$$

Add the two equations. The B -terms cancel:

$$-26A = -26.$$

Divide by -26 :

$$A = 1.$$

Substitute $A = 1$ into the first coefficient equation:

$$-4(1) + 6B = -16.$$

Simplify:

$$-4 + 6B = -16.$$

Starting from $-4 + 6B = -16$, add 4 to both sides:

$$6B = -12.$$

Divide by 6:

$$B = -2.$$

Step 6: Write the particular and general solutions. Insert $A = 1$ and $B = -2$ into the trial:

$$y_p = \cos(3x) - 2 \sin(3x).$$

Combine with y_h :

$$y = e^{-x} (c_1 \cos(2x) + c_2 \sin(2x)) + \cos(3x) - 2 \sin(3x).$$

Block 3 Summary

For a nonresonant forcing term:

1. solve for y_h
2. choose the derivative-closed base trial
3. verify that it does not overlap y_h
4. substitute the trial and its derivatives
5. compare coefficients of independent function types
6. determine every trial coefficient
7. write $y = y_h + y_p$

Polynomial powers, exponentials with different exponents, and trigonometric functions with different frequencies must be compared separately.

Product And Sum Forcing Terms

The Goal Of This Block

The goal is to preserve the complete forcing fingerprint when exponentials, polynomials, and trigonometric functions are multiplied or added.

For a product, keep every visible factor. For a sum, build one trial component for each forcing component and then add them.

Exponential Times Polynomial

Solve:

$$y'' - 2y' + 5y = e^x(8x - 4).$$

Step 1: Solve the homogeneous equation. The characteristic equation is:

$$r^2 - 2r + 5 = 0.$$

Rewrite it as:

$$(r - 1)^2 + 4 = 0.$$

The roots are $1 \pm 2i$, so:

$$y_h = e^x (c_1 \cos(2x) + c_2 \sin(2x)).$$

Step 2: Build and check the product trial. The forcing is e^x multiplied by a first-degree polynomial. Preserve both parts:

$$y_p = e^x(Ax + B).$$

Although y_h contains an e^x envelope, its modes also contain $\cos(2x)$ and $\sin(2x)$. Neither $e^x x$ nor e^x alone is a homogeneous mode, so the trial does not overlap y_h .

Step 3: Differentiate the product explicitly. Start from:

$$y_p = e^x(Ax + B).$$

Apply the product rule once:

$$\begin{aligned} y_p' &= e^x(Ax + B) + e^x(A) \\ &= e^x(Ax + A + B). \end{aligned}$$

Differentiate again:

$$\begin{aligned} y_p'' &= e^x(Ax + A + B) + e^x(A) \\ &= e^x(Ax + 2A + B). \end{aligned}$$

Step 4: Substitute without dropping the common exponential. Use $y_p'' - 2y_p' + 5y_p$:

$$\begin{aligned} y_p'' - 2y_p' + 5y_p &= e^x(Ax + 2A + B) \\ &\quad - 2e^x(Ax + A + B) \\ &\quad + 5e^x(Ax + B). \end{aligned}$$

Factor out e^x :

$$\begin{aligned} y_p'' - 2y_p' + 5y_p &= e^x[(Ax + 2A + B) \\ &\quad - 2(Ax + A + B) + 5(Ax + B)]. \end{aligned}$$

Collect the x -terms and constant terms inside the brackets:

$$\begin{aligned} y_p'' - 2y_p' + 5y_p &= e^x[(A - 2A + 5A)x \\ &\quad + (2A + B - 2A - 2B + 5B)] \\ &= e^x(4Ax + 4B). \end{aligned}$$

Step 5: Compare the polynomial inside e^x . The required forcing is:

$$e^x(8x - 4).$$

Since e^x is nonzero, compare the polynomial coefficients:

$$4Ax + 4B = 8x - 4.$$

Match the x coefficients:

$$4A = 8, \quad A = 2.$$

Match the constant coefficients:

$$4B = -4, \quad B = -1.$$

Thus:

$$y_p = e^x(2x - 1).$$

The general solution is:

$$y = e^x (c_1 \cos(2x) + c_2 \sin(2x)) + e^x(2x - 1).$$

Polynomial Times Trigonometric Forcing

Solve:

$$y'' + y = (-3x + 5) \cos(2x) + (-6x - 1) \sin(2x).$$

Step 1: Find y_h . The homogeneous characteristic equation is:

$$r^2 + 1 = 0.$$

The roots are $\pm i$, so:

$$y_h = c_1 \cos x + c_2 \sin x.$$

Step 2: Build the complete paired trial. The forcing contains first-degree polynomials multiplying sine and cosine at frequency 2. Use:

$$y_p = (Ax + B) \cos(2x) + (Cx + D) \sin(2x).$$

The homogeneous frequency is 1, while the trial frequency is 2. Therefore the trial does not overlap y_h .

Step 3: Differentiate once. Differentiate $(Ax + B) \cos(2x)$ by the product rule:

$$\frac{d}{dx}[(Ax + B) \cos(2x)] = A \cos(2x) - 2(Ax + B) \sin(2x).$$

Differentiate $(Cx + D) \sin(2x)$:

$$\frac{d}{dx}[(Cx + D) \sin(2x)] = C \sin(2x) + 2(Cx + D) \cos(2x).$$

Add $A \cos(2x) - 2(Ax + B) \sin(2x)$ and $C \sin(2x) + 2(Cx + D) \cos(2x)$, then group sine and cosine:

$$y'_p = [A + 2(Cx + D)] \cos(2x) \\ + [C - 2(Ax + B)] \sin(2x).$$

Step 4: Differentiate the grouped first derivative. To keep the product-rule calculation readable, define:

$$P(x) = A + 2(Cx + D), \quad Q(x) = C - 2(Ax + B).$$

Then the grouped first derivative from Step 3 is:

$$y'_p = P(x) \cos(2x) + Q(x) \sin(2x).$$

Differentiate the two coefficient functions:

$$P'(x) = 2C, \quad Q'(x) = -2A.$$

Apply the product rule to the cosine product:

$$\frac{d}{dx} [P(x) \cos(2x)] = P'(x) \cos(2x) - 2P(x) \sin(2x).$$

Apply it separately to the sine product:

$$\frac{d}{dx} [Q(x) \sin(2x)] = Q'(x) \sin(2x) + 2Q(x) \cos(2x).$$

Add the two derivatives and group the cosine and sine terms:

$$y''_p = [P'(x) + 2Q(x)] \cos(2x) \\ + [Q'(x) - 2P(x)] \sin(2x).$$

Now substitute the definitions of P , Q , P' , and Q' . The cosine coefficient becomes:

$$\begin{aligned} P' + 2Q &= 2C + 2[C - 2(Ax + B)] \\ &= 4C - 4(Ax + B). \end{aligned}$$

The sine coefficient becomes:

$$\begin{aligned} Q' - 2P &= -2A - 2[A + 2(Cx + D)] \\ &= -4A - 4(Cx + D). \end{aligned}$$

Therefore:

$$\begin{aligned} y_p'' &= [4C - 4(Ax + B)] \cos(2x) \\ &\quad + [-4A - 4(Cx + D)] \sin(2x). \end{aligned}$$

Step 5: Add y_p to y_p'' . The left side is $y_p'' + y_p$. Add the cosine coefficients:

$$4C - 4(Ax + B) + (Ax + B) = 4C - 3Ax - 3B.$$

Add the sine coefficients:

$$-4A - 4(Cx + D) + (Cx + D) = -4A - 3Cx - 3D.$$

The complete expression for $y_p'' + y_p$ is:

$$\begin{aligned} y_p'' + y_p &= (-3Ax + 4C - 3B) \cos(2x) \\ &\quad + (-3Cx - 4A - 3D) \sin(2x). \end{aligned}$$

Step 6: Compare all four coefficients. Match this expression with:

$$(-3x + 5) \cos(2x) + (-6x - 1) \sin(2x).$$

The $x \cos(2x)$ coefficients give:

$$-3A = -3, \quad A = 1.$$

The $x \sin(2x)$ coefficients give:

$$-3C = -6, \quad C = 2.$$

The constant cosine coefficients give:

$$4C - 3B = 5.$$

Substitute $C = 2$:

$$8 - 3B = 5, \quad B = 1.$$

The constant sine coefficients give:

$$-4A - 3D = -1.$$

Substitute $A = 1$:

$$-4 - 3D = -1, \quad D = -1.$$

Step 7: Assemble the solution. The particular solution is:

$$y_p = (x + 1) \cos(2x) + (2x - 1) \sin(2x).$$

Therefore:

$$y = c_1 \cos x + c_2 \sin x + (x + 1) \cos(2x) + (2x - 1) \sin(2x).$$

Why Sums Can Be Split

Suppose:

$$L[y] = g_1(x) + g_2(x).$$

If:

$$L[y_{p,1}] = g_1(x)$$

and:

$$L[y_{p,2}] = g_2(x),$$

then linearity gives:

$$\begin{aligned} L[y_{p,1} + y_{p,2}] &= L[y_{p,1}] + L[y_{p,2}] \\ &= g_1(x) + g_2(x). \end{aligned}$$

Because these two operator equations have just been established, a particular solution for the combined forcing is:

$$y_p = y_{p,1} + y_{p,2}.$$

This allows a complicated forcing sum to be separated into manageable components. Each component must be checked separately for overlap with y_h .

Block 4 Summary

For products, preserve every factor in the trial. For sums, construct and add separate trial components. In particular:

$$e^{ax}P_n(x) \rightsquigarrow e^{ax}Q_n(x),$$

and:

$$P_n(x) \cos(\omega x) \rightsquigarrow Q_n(x) \cos(\omega x) + R_n(x) \sin(\omega x),$$

where Q_n and R_n are complete degree- n polynomials with undetermined coefficients.

Resonance And The Overlap Correction

The Goal Of This Block

The goal is to repair a base trial that is already part of the homogeneous solution.

The correction rule is:

multiply the complete base trial by x^s

where s is the multiplicity with which the relevant characteristic root or conjugate pair occurs.

This is not a decorative change. It moves the trial out of the homogeneous solution space so the operator can produce a nonzero forcing.

Why An Overlapping Trial Cannot Work

Suppose the forcing is e^{ax} and the base trial is:

$$y_p = Ae^{ax}.$$

If e^{ax} already belongs to y_h , then it satisfies the homogeneous equation:

$$L[e^{ax}] = 0.$$

Multiplying by a constant does not change that output:

$$L[Ae^{ax}] = A L[e^{ax}] = 0.$$

A particular solution for the nonzero forcing e^{ax} must instead satisfy:

$$L[y_p] = e^{ax},$$

which is nonzero. No value of A can make the overlapping trial work.

The practical meaning:

check overlap before differentiating; otherwise an impossible coefficient equation such as $0 = 1$ will appear only after unnecessary work.

Reading The Correcting Power From Multiplicity

The purpose of the overlap check is to answer one question:

how many times must the complete base trial be multiplied by x before it is independent of the homogeneous solution?

The answer is controlled by a characteristic-root multiplicity. Keep the following three roles separate:

1. the **forcing** supplies the exponential rate a and frequency ω
2. the **characteristic polynomial** tells us whether the corresponding roots occur and how many times they occur
3. that multiplicity becomes the correcting power s

Step 1: Read a and ω from the forcing. A general exponential-trigonometric forcing has the form:

$$e^{ax} [P_n(x) \cos(\omega x) + Q_n(x) \sin(\omega x)].$$

Read the two values directly from its visible factors:

- a is the coefficient of x in the exponential e^{ax}
- ω is the frequency in $\cos(\omega x)$ and $\sin(\omega x)$

If no exponential is visible, then $a = 0$ because $e^{0x} = 1$. If no trigonometric factor is visible, then $\omega = 0$.

The polynomial degrees in P_n and Q_n determine the polynomial part of the base trial. They do **not** determine the correcting power.

Step 2: Identify the values to compare with the characteristic roots. The exponential rate and trigonometric frequency combine as:

$$a + i\omega \quad \text{and} \quad a - i\omega.$$

This pairing comes from Euler's formula:

$$e^{(a+i\omega)x} = e^{ax} [\cos(\omega x) + i \sin(\omega x)].$$

We are not replacing the real forcing with a complex one. Euler's formula only tells us which characteristic roots correspond to the real functions $e^{ax} \cos(\omega x)$ and $e^{ax} \sin(\omega x)$.

When $\omega = 0$, both values reduce to the single real value a . A pure polynomial has $a = 0$ and $\omega = 0$, so its comparison value is 0.

Step 3: Read s from the characteristic polynomial. Factor the characteristic polynomial $p(r)$ and compare its roots with $a \pm i\omega$.

- if neither value is a root, set $s = 0$
- if the matching root or pair is simple, set $s = 1$
- if the matching root or pair has multiplicity 2, set $s = 2$
- in general, set s equal to the matching multiplicity

For a real exponential mode, multiplicity s appears in a factor of the form:

$$p(r) = (r - a)^s q(r), \quad q(a) \neq 0.$$

For an exponential-trigonometric mode in a real-coefficient equation, the conjugate roots have the same multiplicity. Their contribution can be written as:

$$p(r) = [(r - a)^2 + \omega^2]^s q(r).$$

If the relevant factor is absent, its multiplicity is zero.

Step 4: Correct the complete base trial. Let $Y_{\text{base}}(x)$ denote the complete trial suggested by the forcing. The corrected trial is:

$$y_p = x^s Y_{\text{base}}(x).$$

The resulting rule is shown below. The example column displays the relevant factor pattern inside $p(r)$; it does not introduce a new forcing term.

Match	Example factor in $p(r)$	s	Correction
no match	relevant factor absent	0	$x^0 = 1$
simple real root	$r - a$	1	x
simple conjugate pair	$(r - a)^2 + \omega^2$	1	x
double real root	$(r - a)^2$	2	x^2
double conjugate pair	$[(r - a)^2 + \omega^2]^2$	2	x^2
multiplicity s	$(\text{relevant factor})^s$	s	x^s

Why does the multiplicity determine the power? If a is a real root of multiplicity s , the homogeneous solution already contains:

$$e^{ax}, \quad xe^{ax}, \quad \dots, \quad x^{s-1}e^{ax}.$$

Multiplying the base trial by x^s moves it beyond every one of these overlapping functions. The same principle applies to a repeated conjugate pair, with both the cosine and sine partners multiplied by the complete factor x^s .

The main point:

read a and ω from the forcing, read s from the characteristic roots, then multiply the complete base trial by x^s .

Repeated Real Resonance

Solve:

$$y'' - 4y' + 4y = e^{2x}.$$

Step 1: Find y_h and the root multiplicity. The characteristic equation is:

$$r^2 - 4r + 4 = 0.$$

Factor:

$$(r - 2)^2 = 0.$$

The root $r = 2$ has multiplicity 2, so:

$$y_h = (c_1 + c_2x)e^{2x}.$$

Step 2: Build and correct the base trial. The forcing e^{2x} suggests the base trial:

$$Ae^{2x}.$$

That function overlaps y_h . Because the root 2 has multiplicity 2, multiply the complete base trial by x^2 :

$$y_p = Ax^2e^{2x}.$$

Step 3: Use the repeated-factor operator efficiently. The differential operator factors as:

$$L = (D - 2)^2,$$

where $D = d/dx$.

Write the corrected trial as:

$$y_p = e^{2x}v(x), \quad v(x) = Ax^2.$$

From Chapter 10:

$$(D - 2)^2[e^{2x}v] = e^{2x}v''.$$

Differentiate $v = Ax^2$ twice:

$$v' = 2Ax, \quad v'' = 2A.$$

Therefore:

$$L[y_p] = 2Ae^{2x}.$$

Step 4: Match the forcing. The differential equation requires:

$$2Ae^{2x} = e^{2x}.$$

Divide by the nonzero function e^{2x} :

$$2A = 1.$$

The equation $2A = 1$ gives:

$$A = \frac{1}{2}.$$

Substitute $A = 1/2$ into the corrected trial $y_p = Ax^2e^{2x}$:

$$y_p = \frac{1}{2}x^2e^{2x}.$$

The general solution is:

$$y = e^{2x} \left(c_1 + c_2x + \frac{1}{2}x^2 \right).$$

Trigonometric Resonance

Solve:

$$y'' + 4y = 6 \cos(2x).$$

Step 1: Find the homogeneous frequency. The characteristic equation is:

$$r^2 + 4 = 0.$$

The roots are $\pm 2i$, so:

$$y_h = c_1 \cos(2x) + c_2 \sin(2x).$$

Step 2: Build and correct the paired trial. Differentiating a cosine produces a sine, and differentiating a sine produces a cosine. The base trial must therefore contain both functions:

$$y_{p,\text{base}} = A \cos(2x) + B \sin(2x).$$

However, both functions in this base trial already occur in y_h . The roots $\pm 2i$ form a simple conjugate pair, so the correcting power is $s = 1$. Multiply the complete paired trial by x :

$$y_p = x[A \cos(2x) + B \sin(2x)].$$

Expand the corrected trial before differentiating:

$$y_p = Ax \cos(2x) + Bx \sin(2x).$$

Step 3: Find the first derivative. Apply the product rule to both terms:

$$\begin{aligned} y_p' &= A [\cos(2x) - 2x \sin(2x)] + B [\sin(2x) + 2x \cos(2x)] \\ &= A \cos(2x) - 2Ax \sin(2x) + B \sin(2x) + 2Bx \cos(2x). \end{aligned}$$

Group the cosine and sine terms:

$$y_p' = (A + 2Bx) \cos(2x) + (B - 2Ax) \sin(2x).$$

Step 4: Find the second derivative. Differentiate the cosine term in y_p' :

$$\frac{d}{dx} [(A + 2Bx) \cos(2x)] = 2B \cos(2x) - 2(A + 2Bx) \sin(2x).$$

Differentiate the sine term in y_p' :

$$\frac{d}{dx} [(B - 2Ax) \sin(2x)] = -2A \sin(2x) + 2(B - 2Ax) \cos(2x).$$

Add these derivatives and group like trigonometric terms:

$$\begin{aligned} y_p'' &= [2B + 2(B - 2Ax)] \cos(2x) \\ &\quad + [-2(A + 2Bx) - 2A] \sin(2x) \\ &= (4B - 4Ax) \cos(2x) + (-4A - 4Bx) \sin(2x). \end{aligned}$$

Step 5: Substitute the complete trial into the differential equation. The left side of the equation is $y_p'' + 4y_p$. From the corrected trial:

$$\begin{aligned} 4y_p &= 4 [Ax \cos(2x) + Bx \sin(2x)] \\ &= 4Ax \cos(2x) + 4Bx \sin(2x). \end{aligned}$$

Now add y_p'' and $4y_p$:

$$\begin{aligned}
 y_p'' + 4y_p &= (4B - 4Ax) \cos(2x) + (-4A - 4Bx) \sin(2x) \\
 &\quad + 4Ax \cos(2x) + 4Bx \sin(2x) \\
 &= (4B - 4Ax + 4Ax) \cos(2x) \\
 &\quad + (-4A - 4Bx + 4Bx) \sin(2x) \\
 &= 4B \cos(2x) - 4A \sin(2x).
 \end{aligned}$$

The $x \cos(2x)$ terms cancel each other, as do the $x \sin(2x)$ terms.

The right side can be written as:

$$6 \cos(2x) + 0 \sin(2x).$$

Match the cosine coefficients and then the sine coefficients:

$$\begin{aligned}
 4B &= 6, \\
 -4A &= 0.
 \end{aligned}$$

Solve for A and B :

$$\begin{aligned}
 B &= \frac{3}{2}, \\
 A &= 0.
 \end{aligned}$$

Substitute these values into the corrected trial:

$$\begin{aligned}
 y_p &= x \left[0 \cos(2x) + \frac{3}{2} \sin(2x) \right] \\
 &= \frac{3}{2} x \sin(2x).
 \end{aligned}$$

Step 6: Verify the particular solution. Differentiate $y_p = \frac{3}{2}x \sin(2x)$:

$$\begin{aligned}y'_p &= \frac{3}{2} \sin(2x) + 3x \cos(2x), \\y''_p &= 3 \cos(2x) + 3 \cos(2x) - 6x \sin(2x) \\&= 6 \cos(2x) - 6x \sin(2x).\end{aligned}$$

Also:

$$4y_p = 6x \sin(2x).$$

Therefore:

$$\begin{aligned}y''_p + 4y_p &= 6 \cos(2x) - 6x \sin(2x) + 6x \sin(2x) \\&= 6 \cos(2x),\end{aligned}$$

which matches the forcing.

Add y_p to the homogeneous solution:

$$y = c_1 \cos(2x) + c_2 \sin(2x) + \frac{3}{2}x \sin(2x).$$

Polynomial Resonance At The Zero Root

Solve:

$$y''' - y'' = 12x - 10.$$

Step 1: Inspect the zero-root multiplicity. The characteristic equation is:

$$r^3 - r^2 = 0.$$

Factor:

$$r^2(r - 1) = 0.$$

The root 0 has multiplicity 2, and the root 1 is simple. Therefore:

$$y_h = c_1 + c_2x + c_3e^x.$$

Step 2: Correct the polynomial base trial. The forcing is first-degree, so the base trial is:

$$Ax + B.$$

Both 1 and x occur in y_h because the zero root has multiplicity 2. Multiply the entire base trial by x^2 :

$$\begin{aligned} y_p &= x^2(Ax + B) \\ &= Ax^3 + Bx^2. \end{aligned}$$

Step 3: Differentiate to the required order. Starting from $y_p = Ax^3 + Bx^2$:

$$\begin{aligned} y_p' &= 3Ax^2 + 2Bx, \\ y_p'' &= 6Ax + 2B, \\ y_p''' &= 6A. \end{aligned}$$

Step 4: Substitute and compare coefficients. The left side is $y_p''' - y_p''$:

$$\begin{aligned}y_p''' - y_p'' &= 6A - (6Ax + 2B) \\ &= (-6A)x + (6A - 2B).\end{aligned}$$

Match this with $12x - 10$:

$$-6A = 12, \quad 6A - 2B = -10.$$

The first coefficient equation is $-6A = 12$. Divide by -6 :

$$A = -2.$$

Substitute $A = -2$ into the second equation:

$$6(-2) - 2B = -10.$$

Simplify:

$$-12 - 2B = -10.$$

Add 12:

$$-2B = 2.$$

The equation $-2B = 2$ gives:

$$B = -1.$$

Substitute $A = -2$ and $B = -1$ into $y_p = Ax^3 + Bx^2$:

$$y_p = -2x^3 - x^2.$$

The complete solution is:

$$y = c_1 + c_2x + c_3e^x - 2x^3 - x^2.$$

Resonance Checklist

Before accepting a trial:

1. solve the characteristic equation
2. write enough of y_h to see every mode
3. build the complete base trial from the forcing
4. identify the forcing exponent and frequency
5. compare them with the characteristic roots
6. read the relevant multiplicity s
7. multiply the entire base trial by x^s
8. only then differentiate and determine coefficients

Do not multiply by x merely because the equation is nonhomogeneous. The correction is used only when overlap occurs, and its power comes from multiplicity.

Block 5 Summary

A base trial that overlaps y_h is annihilated by the homogeneous operator and cannot generate the forcing. If the matching root or conjugate pair has multiplicity s , use:

$$y_p = x^s(\text{complete base trial}).$$

Polynomial resonance is overlap at the zero root. Trigonometric resonance is overlap at a conjugate pair. The same multiplicity rule governs every case.

Complete Problems With Several Moving Parts

The Goal Of This Block

The goal is to coordinate homogeneous solving, trial selection, resonance, sum splitting, and initial conditions without losing the source of any term.

Mixed Polynomial And Resonant Exponential Forcing

Solve:

$$y'' - y' - 2y = 8x - 3 + 5e^{-x}.$$

Step 1: Solve the associated homogeneous equation. The characteristic equation is:

$$r^2 - r - 2 = 0.$$

Factor the characteristic equation $r^2 - r - 2 = 0$:

$$(r - 2)(r + 1) = 0.$$

The roots are 2 and -1 , so:

$$y_h = c_1 e^{2x} + c_2 e^{-x}.$$

Step 2: Split the forcing and build two trials. Write:

$$g_1(x) = 8x - 3, \quad g_2(x) = 5e^{-x}.$$

For the polynomial component, use:

$$y_{p,1} = Ax + B.$$

The zero root is not a characteristic root, so this polynomial trial does not overlap y_h .

For the exponential component, the base trial would be Ce^{-x} . However, e^{-x} occurs in y_h because -1 is a simple root. Multiply that trial by x :

$$y_{p,2} = Cxe^{-x}.$$

The complete corrected trial is:

$$y_p = y_{p,1} + y_{p,2} = Ax + B + Cxe^{-x}.$$

Step 3: Solve the polynomial component locally. For $y_{p,1} = Ax + B$:

$$y'_{p,1} = A, \quad y''_{p,1} = 0.$$

Substitute into $L[y] = y'' - y' - 2y$:

$$\begin{aligned} L[y_{p,1}] &= 0 - A - 2(Ax + B) \\ &= (-2A)x + (-A - 2B). \end{aligned}$$

Match this with $g_1(x) = 8x - 3$:

$$-2A = 8, \quad -A - 2B = -3.$$

The first coefficient equation is $-2A = 8$, so:

$$A = -4.$$

Substitute $A = -4$ into the second equation:

$$-(-4) - 2B = -3.$$

Simplify:

$$4 - 2B = -3.$$

Subtract 4:

$$-2B = -7.$$

The equation $-2B = -7$ gives:

$$B = \frac{7}{2}.$$

Substitute $A = -4$ and $B = 7/2$ into $y_{p,1} = Ax + B$:

$$y_{p,1} = -4x + \frac{7}{2}.$$

Step 4: Solve the resonant exponential component locally. For the corrected trial $y_{p,2} = Cxe^{-x}$, differentiate by the product rule:

$$\begin{aligned} y'_{p,2} &= C[e^{-x} - xe^{-x}] \\ &= C(1 - x)e^{-x}. \end{aligned}$$

Differentiate $y'_{p,2} = C(1-x)e^{-x}$ again:

$$\begin{aligned} y''_{p,2} &= C[-e^{-x} - (1-x)e^{-x}] \\ &= C(x-2)e^{-x}. \end{aligned}$$

Substitute into $L[y] = y'' - y' - 2y$:

$$\begin{aligned} L[y_{p,2}] &= C(x-2)e^{-x} - C(1-x)e^{-x} - 2Cxe^{-x} \\ &= C[(x-2) - (1-x) - 2x]e^{-x} \\ &= C[x-2-1+x-2x]e^{-x} \\ &= -3Ce^{-x}. \end{aligned}$$

Match this with $g_2(x) = 5e^{-x}$:

$$-3Ce^{-x} = 5e^{-x}.$$

Starting from $-3Ce^{-x} = 5e^{-x}$, divide by the nonzero function e^{-x} :

$$-3C = 5.$$

The equation $-3C = 5$ gives:

$$C = -\frac{5}{3}.$$

Substitute $C = -5/3$ into $y_{p,2} = Cxe^{-x}$:

$$y_{p,2} = -\frac{5}{3}xe^{-x}.$$

Step 5: Combine all three parts. The particular solution is:

$$y_p = -4x + \frac{7}{2} - \frac{5}{3}xe^{-x}.$$

Add it to the homogeneous solution:

$$y = c_1e^{2x} + c_2e^{-x} - 4x + \frac{7}{2} - \frac{5}{3}xe^{-x}.$$

The term c_2e^{-x} cannot absorb xe^{-x} because they are linearly independent functions. The resonance correction created a genuinely new particular mode.

A Resonant Initial-Value Problem

Solve:

$$y'' + y = 4 \sin x,$$

subject to:

$$y(0) = 1, \quad y'(0) = 0.$$

Step 1: Find y_h . The characteristic equation is:

$$r^2 + 1 = 0.$$

The roots are $\pm i$, so:

$$y_h = c_1 \cos x + c_2 \sin x.$$

Step 2: Build the corrected trial. The base trial for $4 \sin x$ is:

$$A \cos x + B \sin x.$$

Both modes occur in y_h . The pair $\pm i$ is simple, so multiply the entire trial by x :

$$y_p = x(A \cos x + B \sin x).$$

For this sine forcing, the coefficient calculation will select the cosine component. The two local operator outputs are:

$$\begin{aligned} \left(\frac{d^2}{dx^2} + 1 \right) [x \cos x] &= -2 \sin x, \\ \left(\frac{d^2}{dx^2} + 1 \right) [x \sin x] &= 2 \cos x. \end{aligned}$$

Apply $L = d^2/dx^2 + 1$ to $y_p = x[A \cos x + B \sin x]$ using these two identities:

$$y_p'' + y_p = -2A \sin x + 2B \cos x.$$

Match with $4 \sin x + 0 \cos x$:

$$-2A = 4, \quad 2B = 0.$$

Thus:

$$A = -2, \quad B = 0.$$

The particular solution is:

$$y_p = -2x \cos x.$$

Step 3: Write the general solution before using conditions. Combine y_h and y_p :

$$y = c_1 \cos x + c_2 \sin x - 2x \cos x.$$

Step 4: Apply $y(0) = 1$. Substitute $x = 0$ into the complete solution:

$$1 = c_1 \cos 0 + c_2 \sin 0 - 2(0) \cos 0.$$

Use $\cos 0 = 1$ and $\sin 0 = 0$:

$$1 = c_1.$$

The condition has reduced to $1 = c_1$, so:

$$c_1 = 1.$$

Step 5: Differentiate the complete solution. Differentiate every term in $y = c_1 \cos x + c_2 \sin x - 2x \cos x$:

$$\begin{aligned} y' &= -c_1 \sin x + c_2 \cos x \\ &\quad - 2 \cos x + 2x \sin x. \end{aligned}$$

Step 6: Apply $y'(0) = 0$. Substitute $x = 0$ into the derivative formula:

$$0 = -c_1 \sin 0 + c_2 \cos 0 - 2 \cos 0 + 2(0) \sin 0.$$

Use $\sin 0 = 0$ and $\cos 0 = 1$:

$$0 = c_2 - 2.$$

Add 2:

$$c_2 = 2.$$

Step 7: Write the selected solution. Insert $c_1 = 1$ and $c_2 = 2$:

$$\boxed{y = \cos x + 2 \sin x - 2x \cos x}.$$

Block 6 Summary

For a forcing sum, split the particular solution into matching components and check each component independently for resonance. After all trial coefficients are fixed, reconstruct:

$$y = y_h + y_p.$$

Only then apply initial or boundary conditions to the arbitrary constants in y_h .

A Reliable Workflow, Limits, And Common Mistakes

The Complete Workflow

Step 1: Check that the equation fits the method. Confirm that the equation is linear and has constant coefficients.

Step 2: Isolate the forcing. Write the equation in the form:

$$L[y] = g(x).$$

Step 3: Solve the associated homogeneous equation. Find the characteristic roots and write the complete complementary solution y_h .

Step 4: Split a forcing sum. If $g(x)$ is a sum of manageable components, plan one trial component for each one.

Step 5: Build each base trial. Use the derivative-closed family matching the forcing fingerprint.

Step 6: Perform the overlap check. Compare each complete base trial with y_h . If its associated root has multiplicity s , multiply that entire trial component by x^s .

Step 7: Differentiate the corrected trial. Keep product-rule and chain-rule steps visible, especially for exponential or trigonometric products.

Step 8: Substitute and group like functions. Collect independent types separately: polynomial powers, distinct exponentials, and sine-cosine pairs at each frequency.

Step 9: Solve for the trial coefficients. These coefficients are determined numbers, not arbitrary integration constants.

Step 10: Reconstruct and verify. Write:

$$y = y_h + y_p$$

and verify at least the particular part by checking $L[y_p] = g(x)$.

Step 11: Apply extra conditions last. Use initial or boundary conditions only after the complete general solution is available.

When The Method Applies

The method is designed for linear constant-coefficient equations whose forcing belongs to a finite derivative-closed family.

Typical admissible forcing components include:

$$P_n(x), \quad e^{ax}P_n(x),$$

A third admissible family is:

$$e^{ax} [P_n(x) \cos(\omega x) + Q_n(x) \sin(\omega x)] .$$

Finite sums of these forms are also admissible.

The word **finite** matters. Repeated differentiation must remain inside a finite-dimensional family that can be represented by finitely many undetermined coefficients.

When Another Method Is Needed

Consider:

$$y'' + y = \ln x.$$

Repeated derivatives of $\ln x$ produce:

$$\frac{1}{x}, \quad -\frac{1}{x^2}, \quad \frac{2}{x^3}, \quad \dots$$

No finite trial family closes under differentiation, so undetermined coefficients is not the appropriate method.

Now consider:

$$xy'' + y = e^x.$$

The coefficient of y'' is the variable x , not a constant. Even though the forcing is an exponential, the standard constant-coefficient trial rules do not apply.

Variation of parameters, introduced in Chapter 12, handles a broader class of linear equations.

Common Mistake: Choosing A Trial Before Finding y_h

For:

$$y'' - 6y' + 9y = e^{3x},$$

the forcing suggests Ae^{3x} . However, the characteristic equation is:

$$(r - 3)^2 = 0.$$

Thus e^{3x} and xe^{3x} already occur in y_h . The corrected trial must be:

$$Ax^2e^{3x}.$$

Finding y_h first makes the required power of x visible.

Common Mistake: Omitting Lower Polynomial Powers

For a cubic forcing, the trial must be:

$$Ax^3 + Bx^2 + Cx + D.$$

Using only Ax^3 assumes in advance that the lower coefficients are zero. The differential operator may require nonzero lower-degree terms even when some powers are absent from the forcing.

Common Mistake: Using Only The Visible Trigonometric Function

For a forcing proportional to $\sin(4x)$, do not use only $A \sin(4x)$. Differentiation creates cosine, so use:

$$A \cos(4x) + B \sin(4x).$$

The paired family must also be multiplied as a unit if resonance occurs.

Common Mistake: Multiplying Only One Term By x^s

Suppose the corrected base family is:

$$(Ax + B) \cos(2x) + (Cx + D) \sin(2x)$$

and the relevant conjugate pair has multiplicity 2. The correction is:

$$x^2 [(Ax + B) \cos(2x) + (Cx + D) \sin(2x)] .$$

The factor x^2 multiplies the entire trial component. Applying it to only the sine term or only the highest-degree terms destroys the derivative-closed family.

Common Mistake: Treating Trial Coefficients As Arbitrary

In:

$$y_p = Ax^2 + Bx + C,$$

the values A, B, C are determined by substitution. They are not extra integration constants.

The arbitrary constants appear only in y_h . Adding arbitrary constants to y_p would duplicate homogeneous freedom already represented there.

Common Mistake: Reporting Only y_p

The method constructs a particular solution, but the general solution is:

$$y = y_h + y_p.$$

Unless the problem asks only for a particular solution or only for the trial form, stopping at y_p leaves out the arbitrary homogeneous family.

Block 7 Summary

The method succeeds when three decisions are made in the correct order:

1. identify the homogeneous modes
2. build a derivative-closed trial from the forcing
3. correct every overlap using the exact root multiplicity

After substitution determines the trial coefficients, attach the particular solution to y_h and apply any additional conditions.

Original Practice Set

Practice Problems

Problem 1: Build A Complete Polynomial Trial. An equation has constant coefficients, its homogeneous solution contains no polynomial modes, and its forcing is:

$$5x^4 - 3x + 2.$$

Write the correct trial form for y_p .

Problem 2: Preserve An Exponential-Polynomial Product. An equation has no homogeneous mode with exponential rate -2 . Its forcing is:

$$e^{-2x}(x^2 + 4).$$

Write the correct trial form.

Problem 3: Preserve Both Trigonometric Partners. An equation has no homogeneous mode at frequency 4. Its forcing is:

$$(3x - 1) \sin(4x).$$

Write the correct trial form.

Problem 4: Correct A Simple Exponential Resonance. The complementary solution contains $c_1 e^{3x}$, and the forcing is $7e^{3x}$. The root 3 is simple. Write the corrected trial form.

Problem 5: Correct A Repeated Product Resonance. The complementary solution contains:

$$e^{-x}(c_1 + c_2 x),$$

and the forcing is:

$$e^{-x}(2x + 5).$$

Write the corrected trial form.

Problem 6: Recognise Polynomial Resonance. The complementary solution is:

$$y_h = c_1 + c_2 x + c_3 e^{2x},$$

and the forcing is $4x - 1$. Write the corrected trial form.

Problem 7: Solve A Polynomially Forced Equation. Solve:

$$y'' - 3y' + 2y = -2x^2 + 10x - 6.$$

Problem 8: Solve An Exponentially Forced Equation. Solve:

$$y'' - 2y' + 2y = 10e^{-x}.$$

Problem 9: Solve A Trigonometrically Forced Equation. Solve:

$$y'' + 3y' + 2y = -10 \cos(2x) - 10 \sin(2x).$$

Problem 10: Solve A Repeated Resonance. Solve:

$$y'' - 6y' + 9y = 12e^{3x}.$$

Problem 11: Solve A Mixed Forcing Problem. Solve:

$$y'' - y' - 2y = -2x - 5 + 6e^{2x}.$$

Problem 12: Solve A Resonant Initial-Value Problem. Solve:

$$y'' + 4y = 8 \cos(2x),$$

subject to:

$$y(0) = 3, \quad y'(0) = 1.$$

Problem 13: Diagnose The Method's Scope. For each equation, state whether the standard method of undetermined coefficients applies and explain why:

$$(a) \quad y'' + 5y = xe^{-x},$$

$$(b) \quad y'' + y = \ln x,$$

$$(c) \quad xy'' - y' = e^x.$$

Worked Answers: Problems 1 To 6

Answer 1. The forcing is:

$$5x^4 - 3x + 2.$$

Its highest power is x^4 , so use a complete fourth-degree polynomial. The missing x^3 and x^2 terms in the forcing do not remove those terms from the trial:

$$y_p = Ax^4 + Bx^3 + Cx^2 + Dx + E.$$

The problem states that no polynomial homogeneous modes are present, so no power of x is needed for overlap correction.

Answer 2. The forcing is:

$$e^{-2x}(x^2 + 4).$$

Preserve the exponential e^{-2x} and attach a complete quadratic:

$$y_p = e^{-2x}(Ax^2 + Bx + C).$$

The linear term must be included even though it is absent from $x^2 + 4$. The problem states that exponential rate -2 does not overlap the homogeneous solution, so the base trial is unchanged.

Answer 3. The forcing is:

$$(3x - 1) \sin(4x).$$

The polynomial has degree 1, and differentiation exchanges sine and cosine. Therefore both trigonometric partners require complete first-degree polynomials:

$$y_p = (Ax + B) \cos(4x) + (Cx + D) \sin(4x).$$

There is no homogeneous mode at frequency 4, so no resonance correction is required.

Answer 4. The forcing $7e^{3x}$ gives the base trial:

$$Ae^{3x}.$$

However, e^{3x} already occurs in the complementary solution, and root 3 is simple. Multiply the entire base trial by x^1 :

$$y_p = Axe^{3x}.$$

Answer 5. The complementary solution contains:

$$e^{-x}(c_1 + c_2x),$$

so root -1 has multiplicity 2.

The forcing $e^{-x}(2x + 5)$ produces the base trial:

$$e^{-x}(Ax + B).$$

Because exponential rate -1 overlaps with multiplicity 2, multiply the complete base trial by x^2 :

$$\boxed{y_p = x^2 e^{-x}(Ax + B)}.$$

Answer 6. The complementary solution is:

$$y_h = c_1 + c_2 x + c_3 e^{2x}.$$

The modes 1 and x show that the zero root has multiplicity 2.

The first-degree polynomial forcing $4x - 1$ gives the base trial:

$$Ax + B.$$

Multiply that entire trial by x^2 :

$$\boxed{y_p = x^2(Ax + B)}.$$

Equivalently, the corrected trial may be expanded as:

$$y_p = Ax^3 + Bx^2.$$

Worked Answers: Problems 7 To 10

Answer 7. Solve:

$$y'' - 3y' + 2y = -2x^2 + 10x - 6.$$

The associated characteristic equation is:

$$r^2 - 3r + 2 = 0.$$

Factor the characteristic equation $r^2 - 3r + 2 = 0$:

$$(r - 1)(r - 2) = 0.$$

Thus:

$$y_h = c_1 e^x + c_2 e^{2x}.$$

The forcing is quadratic and does not overlap y_h , so use:

$$y_p = Ax^2 + Bx + C.$$

Differentiate:

$$y'_p = 2Ax + B, \quad y''_p = 2A.$$

Substitute into $y''_p - 3y'_p + 2y_p$:

$$\begin{aligned} y''_p - 3y'_p + 2y_p &= 2A - 3(2Ax + B) + 2(Ax^2 + Bx + C) \\ &= (2A)x^2 + (-6A + 2B)x + (2A - 3B + 2C). \end{aligned}$$

Match with $-2x^2 + 10x - 6$:

$$\begin{aligned}2A &= -2, \\ -6A + 2B &= 10, \\ 2A - 3B + 2C &= -6.\end{aligned}$$

The first equation gives:

$$A = -1.$$

Substitute $A = -1$ into the second equation:

$$-6(-1) + 2B = 10.$$

Therefore:

$$6 + 2B = 10, \quad B = 2.$$

Substitute $A = -1$ and $B = 2$ into the third equation:

$$2(-1) - 3(2) + 2C = -6.$$

Simplify:

$$-8 + 2C = -6.$$

Add 8 and divide by 2:

$$C = 1.$$

Substitute $A = -1$, $B = 2$, and $C = 1$ into the quadratic trial:

$$y_p = -x^2 + 2x + 1.$$

The general solution is:

$$y = c_1 e^x + c_2 e^{2x} - x^2 + 2x + 1.$$

Answer 8. Solve:

$$y'' - 2y' + 2y = 10e^{-x}.$$

The associated characteristic equation is:

$$r^2 - 2r + 2 = 0.$$

Rewrite it as:

$$(r - 1)^2 + 1 = 0.$$

The roots are $1 \pm i$, so:

$$y_h = e^x (c_1 \cos x + c_2 \sin x).$$

The forcing exponential rate is -1 , which does not match the homogeneous roots. Use:

$$y_p = Ae^{-x}.$$

Differentiate:

$$y_p' = -Ae^{-x}, \quad y_p'' = Ae^{-x}.$$

Substitute into the left side:

$$\begin{aligned} y_p'' - 2y_p' + 2y_p &= Ae^{-x} - 2(-Ae^{-x}) + 2Ae^{-x} \\ &= 5Ae^{-x}. \end{aligned}$$

Match with $10e^{-x}$:

$$5Ae^{-x} = 10e^{-x}.$$

Divide by e^{-x} and then by 5:

$$A = 2.$$

Therefore:

$$y = e^x(c_1 \cos x + c_2 \sin x) + 2e^{-x}.$$

Answer 9. Solve:

$$y'' + 3y' + 2y = -10 \cos(2x) - 10 \sin(2x).$$

The characteristic equation is:

$$r^2 + 3r + 2 = 0.$$

Factor the characteristic equation $r^2 + 3r + 2 = 0$:

$$(r + 1)(r + 2) = 0.$$

Thus:

$$y_h = c_1 e^{-x} + c_2 e^{-2x}.$$

The nonresonant paired trial is:

$$y_p = A \cos(2x) + B \sin(2x).$$

Differentiate:

$$y'_p = -2A \sin(2x) + 2B \cos(2x),$$

$$y''_p = -4A \cos(2x) - 4B \sin(2x).$$

Substitute and group sine and cosine:

$$\begin{aligned} y''_p + 3y'_p + 2y_p &= (-2A + 6B) \cos(2x) \\ &\quad + (-6A - 2B) \sin(2x). \end{aligned}$$

Match coefficients with the forcing:

$$-2A + 6B = -10,$$

$$-6A - 2B = -10.$$

Divide both equations by -2 :

$$A - 3B = 5,$$

$$3A + B = 5.$$

The first simplified coefficient equation is $A - 3B = 5$. Solve it for A :

$$A = 5 + 3B.$$

Substitute this expression into $3A + B = 5$:

$$3(5 + 3B) + B = 5.$$

Expand and combine:

$$15 + 10B = 5.$$

Subtract 15 and divide by 10:

$$B = -1.$$

Return to $A = 5 + 3B$:

$$A = 5 + 3(-1) = 2.$$

Thus:

$$y_p = 2 \cos(2x) - \sin(2x).$$

The general solution is:

$$y = c_1 e^{-x} + c_2 e^{-2x} + 2 \cos(2x) - \sin(2x).$$

Answer 10. Solve:

$$y'' - 6y' + 9y = 12e^{3x}.$$

The characteristic equation is:

$$r^2 - 6r + 9 = 0.$$

Factor:

$$(r - 3)^2 = 0.$$

Therefore:

$$y_h = e^{3x}(c_1 + c_2x).$$

The base trial Ae^{3x} overlaps a root of multiplicity 2. Correct it by multiplying by x^2 :

$$y_p = Ax^2e^{3x}.$$

The operator is $(D - 3)^2$. Write $y_p = e^{3x}v$ with $v = Ax^2$. Then:

$$(D - 3)^2[e^{3x}v] = e^{3x}v''.$$

Using $v'' = 2A$ in $(D - 3)^2[e^{3x}v] = e^{3x}v''$ gives:

$$y_p'' - 6y_p' + 9y_p = 2Ae^{3x}.$$

Match this with $12e^{3x}$:

$$2A = 12, \quad A = 6.$$

The complete solution is:

$$y = e^{3x}(c_1 + c_2x + 6x^2).$$

Worked Answers: Problems 11 To 13

Answer 11. Solve:

$$y'' - y' - 2y = -2x - 5 + 6e^{2x}.$$

The characteristic equation is:

$$r^2 - r - 2 = 0.$$

Factor:

$$(r - 2)(r + 1) = 0.$$

Therefore:

$$y_h = c_1e^{2x} + c_2e^{-x}.$$

Split the forcing into:

$$g_1(x) = -2x - 5, \quad g_2(x) = 6e^{2x}.$$

For the polynomial component, use:

$$y_{p,1} = Ax + B.$$

Then:

$$y'_{p,1} = A, \quad y''_{p,1} = 0.$$

Substitute into $L[y] = y'' - y' - 2y$:

$$\begin{aligned} L[y_{p,1}] &= 0 - A - 2(Ax + B) \\ &= (-2A)x + (-A - 2B). \end{aligned}$$

Match with $-2x - 5$:

$$-2A = -2, \quad -A - 2B = -5.$$

The first coefficient equation $-2A = -2$ gives $A = 1$. Substitute this value into the second equation $-A - 2B = -5$:

$$-1 - 2B = -5.$$

Add 1:

$$-2B = -4.$$

Thus:

$$B = 2$$

and:

$$y_{p,1} = x + 2.$$

For $6e^{2x}$, the base trial Ce^{2x} overlaps the simple root 2. Multiply by x :

$$y_{p,2} = Cxe^{2x}.$$

Differentiate:

$$y'_{p,2} = C(1 + 2x)e^{2x},$$

$$y''_{p,2} = C(4 + 4x)e^{2x}.$$

Substitute into the operator:

$$\begin{aligned} L[y_{p,2}] &= C(4 + 4x)e^{2x} - C(1 + 2x)e^{2x} - 2Cxe^{2x} \\ &= C[(4 + 4x) - (1 + 2x) - 2x]e^{2x} \\ &= 3Ce^{2x}. \end{aligned}$$

Match with $6e^{2x}$:

$$3C = 6, \quad C = 2.$$

Therefore:

$$y_{p,2} = 2xe^{2x}.$$

Combine y_h , $y_{p,1}$, and $y_{p,2}$:

$$y = c_1 e^{2x} + c_2 e^{-x} + x + 2 + 2x e^{2x}.$$

Answer 12. Solve:

$$y'' + 4y = 8 \cos(2x),$$

with:

$$y(0) = 3, \quad y'(0) = 1.$$

The characteristic equation is:

$$r^2 + 4 = 0.$$

The roots are $\pm 2i$, so:

$$y_h = c_1 \cos(2x) + c_2 \sin(2x).$$

The base trigonometric trial overlaps this simple conjugate pair. Use the corrected trial:

$$y_p = x[A \cos(2x) + B \sin(2x)].$$

The local operator outputs are:

$$\left(\frac{d^2}{dx^2} + 4 \right) [x \cos(2x)] = -4 \sin(2x)$$

and:

$$\left(\frac{d^2}{dx^2} + 4 \right) [x \sin(2x)] = 4 \cos(2x).$$

Apply $L = d^2/dx^2 + 4$ to the corrected trial using these two local operator outputs:

$$y_p'' + 4y_p = 4B \cos(2x) - 4A \sin(2x).$$

Match with $8 \cos(2x) + 0 \sin(2x)$:

$$4B = 8, \quad -4A = 0.$$

Thus:

$$B = 2, \quad A = 0.$$

The general solution before applying the conditions is:

$$y = c_1 \cos(2x) + c_2 \sin(2x) + 2x \sin(2x).$$

Apply $y(0) = 3$:

$$3 = c_1 \cos 0 + c_2 \sin 0 + 2(0) \sin 0.$$

Using $\cos 0 = 1$ and $\sin 0 = 0$, the condition reduces to $3 = c_1$. Therefore:

$$c_1 = 3.$$

Differentiate the complete general solution:

$$\begin{aligned} y' &= -2c_1 \sin(2x) + 2c_2 \cos(2x) \\ &\quad + 2 \sin(2x) + 4x \cos(2x). \end{aligned}$$

Apply $y'(0) = 1$:

$$1 = -2c_1 \sin 0 + 2c_2 \cos 0 + 2 \sin 0 + 4(0) \cos 0.$$

Use $\sin 0 = 0$ and $\cos 0 = 1$:

$$1 = 2c_2.$$

Therefore:

$$c_2 = \frac{1}{2}.$$

The selected solution is:

$$y = 3 \cos(2x) + \frac{1}{2} \sin(2x) + 2x \sin(2x).$$

Answer 13. For part (a):

$$y'' + 5y = xe^{-x}.$$

The equation is linear with constant coefficients. Its forcing is a first-degree polynomial multiplied by an exponential, which belongs to a finite derivative-closed family. The standard method applies. Before using the base trial:

$$e^{-x}(Ax + B),$$

the root -1 must be compared with the characteristic roots of $r^2 + 5 = 0$.

For part (b):

$$y'' + y = \ln x.$$

The equation has constant coefficients, but repeated derivatives of $\ln x$ produce an unending sequence of distinct reciprocal powers. There is no finite standard trial family. Undetermined coefficients does not apply.

For part (c):

$$xy'' - y' = e^x.$$

The coefficient of y'' is the variable x . The equation is linear but does not have constant coefficients, so the standard method does not apply.

Chapter 11 Final Summary

For a linear constant-coefficient equation:

$$L[y] = g(x),$$

the general solution is:

$$y = y_h + y_p.$$

Construct y_h from the associated equation $L[y] = 0$. Construct one y_p by matching a finite derivative-closed trial family to the forcing.

The basic fingerprints are:

$$P_n(x), \quad e^{ax}P_n(x),$$

The exponential-trigonometric fingerprint is:

$$e^{ax} [P_n(x) \cos(\omega x) + Q_n(x) \sin(\omega x)] .$$

For a forcing sum, add matching trial components. Before differentiating, compare every complete base trial with y_h .

If the relevant characteristic root or conjugate pair has multiplicity s , use:

$$y_p = x^s (\text{complete base trial}) .$$

The chapter's final lesson is:

solve the homogeneous problem first, preserve the forcing's complete derivative family, correct overlap by the exact multiplicity, and make every trial coefficient earn its value through substitution.